

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

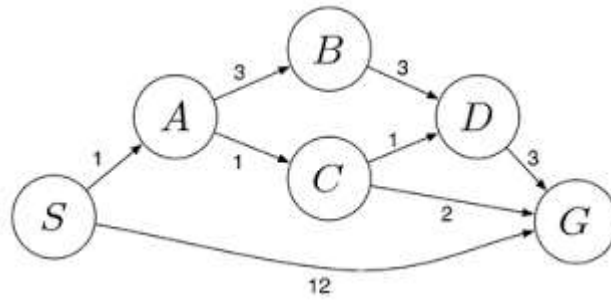
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I Subash G [192572150] (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Subash G*

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

1. Water Jug Problem

Solution

The Water Jug Problem is a state-space search problem in Artificial Intelligence. The objective is to obtain exactly 2 gallons of water in the 4-gallon jug using a 4-gallon jug and a 3-gallon jug.

Step	Action Performed	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug	0	3
2	Pour 3-gallon jug into 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour water into the 4-gallon jug until it becomes full	4	2
5	Empty the 4-gallon jug	0	2
6	Pour the remaining 2 gallons into the 4-gallon jug	2	0

State Space Representation

(0,0),(0,3),(3,0),(3,3),(4,2),(0,2),(2,0)

Goal State

(2,0)

The 4-gallon jug contains exactly 2 gallons of water. Therefore, the goal is achieved.

2. Mars Rover Agent

Percepts of the Agent

Percepts are the information received by the rover through its sensors.

- Images of the surroundings
- Rock and soil composition data
- Atmospheric information
- Terrain conditions
- Obstacle information
- Battery status

ii. Characterization of Operating Environment

The Mars Rover operates in a:

- Partially Observable Environment
- Dynamic Environment
- Sequential Environment

- Uncertain Environment
- Real-World Environment

The rover cannot observe everything at once, and environmental conditions may change continuously.

iii. Actions of the Agent

The rover can:

- Move forward, backward, left, and right
- Avoid obstacles
- Capture images
- Collect soil and rock samples
- Analyze collected samples
- Transmit data to Earth

iv. Performance Measure

The performance of the rover can be evaluated based on:

- Safe navigation
- Obstacle avoidance
- Accuracy of collected data
- Battery efficiency
- Successful completion of mission objectives
- Amount of useful scientific information gathered

v. Suitable Agent Architecture

A Goal-Based Agent with Utility-Based features is most suitable.

Reason

A Goal-Based Agent helps the rover achieve mission objectives.

A Utility-Based Agent helps select the best action considering:

- Safety
- Battery power
- Terrain conditions
- Scientific value of collected samples

Thus, the rover can operate efficiently in an uncertain environment.

3. 8-Queens Problem

Problem Statement

Place 8 queens on a chessboard such that no queen attacks another queen.

Problem Components

Initial State

The chessboard is empty and no queens are placed.

State Space

The state space consists of all possible arrangements of 8 queens on the chessboard.

Actions

Place one queen in a safe position in the next column without attacking any previously placed queen.

Goal Test

The goal is achieved when:

- All 8 queens are placed.
- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are in the same diagonal.

Path Cost:

Each queen placement has equal cost.

Therefore, the path cost is uniform.

Search Strategy Used-Backtracking Search (Depth First Search Based)

Working Procedure

1. Place the first queen in the first column.
2. Move to the next column and place another queen in a safe position.
3. Continue placing queens column by column.
4. If no safe position is available, backtrack to the previous column.
5. Try another position.
6. Repeat until all 8 queens are placed safely.

State Representation

Initial State:

Empty Chessboard

Intermediate State:

Some queens placed safely.

Goal State:

Eight queens placed without attacking each other.

Conclusion

The 8-Queens problem demonstrates the use of Backtracking Search and Depth First Search for solving Constraint Satisfaction Problems efficiently.

4. OLA Cab Booking Application

Type of Problem-Solving Agent

Goal-Based Agent

Reason

The goal of the system is to help the customer reach the destination safely and comfortably.

The agent considers:

- Pickup location
- Destination
- Preferred cab type
- Cab availability

before making a decision.

Pseudocode

Start

Input Pickup Location

Input Destination

Display Available Cab Types

(Mini, Micro, Sedan, Prime, Shared)

Customer Selects Cab Type
Check Cab Availability
If Cab Available
Calculate Fare
Assign Nearest Driver
Confirm Booking
Display Driver and Trip Details
Else
Display "Cab Not Available"
Suggest Alternative Cab Type
End If
End

Working

1. Customer enters pickup location.
2. Customer enters destination.
3. Available cab types are displayed.
4. Customer selects a preferred cab.
5. System checks availability.
6. If available, fare is calculated.
7. Nearest driver is assigned.
8. Booking is confirmed.
9. If unavailable, another cab option is suggested.

A Goal-Based Agent is suitable because it focuses on achieving the customer's objective of reaching the destination efficiently and comfortably.

5. Uniform Cost Search (UCS)

Uniform Cost Search is an uninformed search algorithm that expands the node with the lowest cumulative path cost first. The objective is to determine the least-cost path from the starting warehouse S to the destination warehouse G.

Given Edge Costs

$$S \rightarrow A = 1$$

$$S \rightarrow G = 12$$

$$A \rightarrow B = 3$$

$$A \rightarrow C = 1$$

$$B \rightarrow D = 3$$

$$C \rightarrow D = 1$$

$$C \rightarrow G = 2$$

$$D \rightarrow G = 3$$

UCS Process

Step 1

Start Node:

S

Cost = 0

Expand S

Generated Paths:

$$S \rightarrow A = 1$$

$$S \rightarrow G = 12$$

Choose A because it has lower cost.

Step 2

Expand A

Generated Paths:

$$S \rightarrow A \rightarrow B = 4, S \rightarrow A \rightarrow C = 2$$

Choose C because it has lower cost.

Step 3

Expand C

Generated Paths:

$$S \rightarrow A \rightarrow C \rightarrow D = 3, S \rightarrow A \rightarrow C \rightarrow G = 4$$

Choose D because it has lower cost.

Step 4

Expand D

Generated Path:

$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 6$

A goal node G is already available with cost 4.

Therefore, UCS selects the lower-cost goal node.

Path Cost Comparison

Path	Total Cost
$S \rightarrow G$	12
$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$	10
$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	6
$S \rightarrow A \rightarrow C \rightarrow G$	4

Least-Cost Path

Optimal Path:

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost:

$$1 + 1 + 2 = 4$$

Using Uniform Cost Search, the least-cost path from S to G is:

$S \rightarrow A \rightarrow C \rightarrow G$

Minimum Cost = 4.

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